GSE Charging and Energy Management for Electric Airport Operations

As electric GSE fleets grow, charging becomes part of daily airport operations.

Airports need to keep electric ground support equipment available, avoid charging queues, and manage limited power capacity across gates, stands, and service areas.

This page explains how coordinated charging and energy management can help airports scale electric GSE without creating new operational bottlenecks.

Based on ITW GSE's experience with gate equipment and electric GSE charging

We work with ground power, pre-conditioned air, charging and power management for airport operations, and we have done so for decades.

This page explains how charging can be planned as part of daily gate and apron operations using solutions such as **Power Share**, **EcoGate**, **DC Charger**, and **Intelligent Power Management**.

OUR HISTORY
OUR TEAM
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How do we avoid GSE charging bottlenecks?

GSE charging bottlenecks can be reduced by placing charging closer to operations and coordinating charging with available power capacity.

Charging bottlenecks often occur when too many units compete for the same charging infrastructure at the same time. This can create waiting time, congestion and reduced GSE availability.

To reduce charging bottlenecks, airports typically focus on gate-level charging, opportunity charging during idle periods, and coordinated power usage across charging, GPU and PCA.

ITW GSE supports this approach through solutions such as **Power Share** and integrated charging concepts within **EcoGate**.

SEE HOW POWER SHARE SUPPORTS GSE CHARGING

EXPLORE INTEGRATED GATE OPERATIONS WITH ECOGATE

Overview of charging and energy management challenges

Challenge	What it means in practice	What to consider first
Charging bottlenecks	Too many GSE units competing for the same chargers during peak periods	Move charging closer to operations and use opportunity charging
Limited gate power capacity	GPUs, PCA and charging may compete for available gate-level power	Prioritize aircraft-critical systems and coordinate loads
Simultaneous peak demand	Charging several units simultaneously can increase peak load and overload risk	Coordinate charging with operational demand and idle perio
Mixed gate and remote stand operations	Charging infrastructure may not support all operational areas equally	Combine fixed charging with flexible and mobile charging setup

alignment to operational workflows

scale charging progressively

Can charging be integrated into gate operations?

Yes. GSE charging can be integrated into gate operations when charging, GPU, and PCA are coordinated as one system.

At many airports, aircraft support systems and GSE charging depend on the same available power. If these loads are not coordinated, charging can compete with turnaround-critical functions.

A coordinated gate setup allows aircraft power and PCA to remain the priority, while charging uses available capacity when operational demand allows.

The ITW GSE EcoGate concept is designed around this type of system-based gate operation.

EXPLORE INTEGRATED GATE OPERATIONS WITH ECOGATE

VIEW DC CHARGING SOLUTIONS FOR ELECTRIC GSE

SEE HOW POWER SHARE SUPPORTS GSE CHARGING

Why is GSE charging becoming an operational challenge?

Electric GSE fleets create new power demand across the airport, and the challenge is not only installing chargers.

It is making sure equipment is charged where and when it is needed, without creating queues, downtime, or unnecessary towing across the apron.

Working with GSE charging, airports need to consider:

- Where charging should take place

- How charging affects turnaround operations

Without a clear charging strategy, electrification can create new operational bottlenecks instead of reducing them.

VIEW DC CHARGING SOLUTIONS FOR ELECTRIC GSE

SEE HOW POWER SHARE SUPPORTS GSE CHARGING



How can airports use existing power more efficiently?



Airports can use existing power more efficiently by coordinating loads and sharing available capacity between connected equipment.

In many cases, the challenge is not only total grid capacity — but how power is used during daily operations.

To improve energy utilization, airports often focus on:

- Avoiding simultaneous peak demand
- Prioritizing aircraft-critical systems
- Sharing unused capacity between equipment
- Coordinating charging with operational demand

ITW GSE Power Share supports this approach by enabling available gate power to be used for charging electric GSE when capacity is available.

SEE HOW POWER SHARE SUPPORTS GSE CHARGING

When and where should GSE be charged?

GSE should be charged where it creates the least operational disruption and where the equipment is already operating.

For some airports, centralized charging will remain useful for planned or overnight charging. For others, gate-level or remote stand charging can reduce towing, improve availability and support opportunity charging during natural idle periods.

The right setup depends on traffic flow, available power, gate layout and how the equipment is used throughout the day.

Feel free to reach out to our **Sales team** for a chat about your needs for charging.

Which charging setup fits your airport operation?

Situation	What to consider	Relevant ITW GSE solution
Too much equipment is waiting for chargers	Move charging closer to daily operations	Power Share EcoGate
Gate power is shared between GPU, PCA and charging	Coordinate and prioritize available power	Intelligent Power Manage Power Share
Equipment is charged far from where it is used	Reduce towing and support opportunity charging	DC Charger
The airport has both gates and remote stands	Combine fixed and flexible charging options	DC Charger Power Share

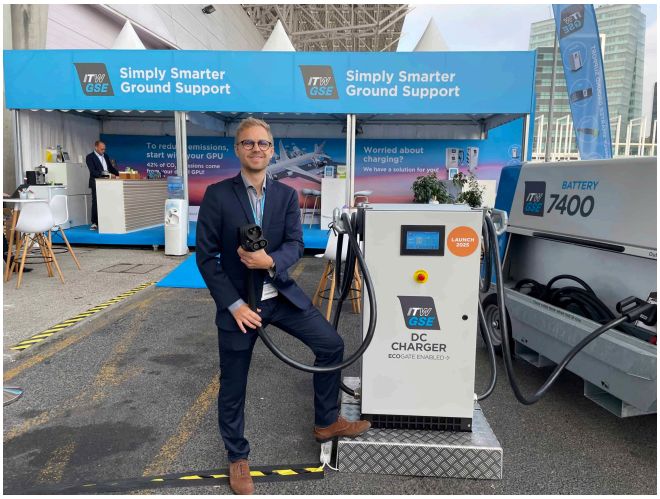
How do charging bottlenecks affect GSE availability?

Charging bottlenecks reduce GSE availability because equipment may spend time waiting, towing, or charging away from operations.

This can affect:

- Equipment readiness during peak periods
- Turnaround efficiency
- Apron traffic and congestion
- Overall fleet utilization

A strong charging strategy focuses not only on charger installation — but on





What is the difference between centralized charging and gate-level charging?

Centralized charging places chargers in dedicated charging areas, while gate-level charging places charging closer to daily operations.

Centralized charging can work well for planned or overnight charging. Gate-level charging supports opportunity charging closer to where equipment is used.

Many airports combine both approaches to reduce towing, improve GSE availability and support different operational scenarios.

The right setup depends on traffic flow, infrastructure and operational priorities.

The system approach of EcoGate lets you combine your GSE to suite your many different daily operations.

ECOGATE - A GATE SYSTEM APPROACH

Why does energy management matter

Energy management helps airports coordinate charging demand across gates, stands and operational areas.

As electric GSE fleets grow, simultaneous charging can increase peak load and place pressure on existing infrastructure.

Energy management helps airports:

- Prioritize critical loads
- Reduce overload risk
- Coordinate charging with gate operations
- Use existing infrastructure more efficiently

This becomes increasingly important when charging operates alongside **GPUs**, **PCA** systems and other gate equipment.

The ITW GSE **Intelligent Power Management** (IPM) is a system designed to prioritize the GPU all while the PCA is still running.

LEARN ABOUT INTELLIGENT POWER MANAGEMENT (IPM)

From charging equipment to charging strategy

Scaling electric GSE is not only about installing chargers.

It is about building a charging strategy that supports daily operations.

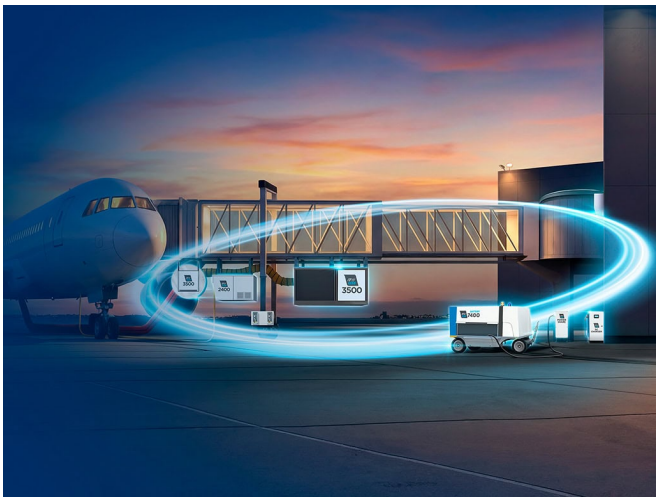
A strong charging strategy considers:

- Where equipment operates
- When equipment is available
- Aircraft support requirements
- Future electrification growth
- Available infrastructure capacity

This helps airports reduce operational pressure while scaling electric GSE in a controlled way.



Plan GSE charging as part of airport operations



Electric GSE charging works best when it is planned as part of the overall gate and apron operation.

Charging should be coordinated with aircraft turnaround, GPU and PCA demand, equipment availability and infrastructure capacity.

A coordinated approach can help airports reduce charging bottlenecks, improve operational reliability and scale electrification more efficiently.

The ITW GSE EcoGate system helps airports evaluate how charging, power sharing and coordinated gate operations can support long-term electrification goals.

GROUND POWER UNITS

PRE-CONDITIONED AIR UNITS

ECOGATE - A GATE SYSTEM APPROACH

How Power Share, EcoGate, and Intelligent Power Management work together

EcoGate is ITW GSE's system approach to smarter gate electrification. Instead of looking at GPUs, PCAs, chargers, cables, and power infrastructure as separate units, EcoGate brings them together in one coordinated setup where power is used where it creates the most operational value.

Power Share and Intelligent Power Management both support this system approach.

Together, they help airports get more out of the power they already have. EcoGate provides the overall system thinking. Intelligent Power Management helps coordinate power use at the gate, while Power Share makes it possible to turn available excess capacity into practical charging opportunities.

POWER SHARE | ECOGATE - A SYSTEM APPROACH | INTELLIGENT POWER MANAGEMENT

Frequently Asked Questions

How do airports avoid GSE charging bottlenecks?

When should GSE be charged?

Can GSE charging be integrated into gate operations?

What is opportunity charging for GSE?

How can airports charge electric GSE without overloading infrastructure?

What role does Power Share play in GSE charging?

What role does EcoGate play in charging and energy management?

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